

AY2025-26

Web Technology

Quality Laboratory Manual

ExperimentNo. 06

Program Based on XSLT Stylesheet to Convert XML to HTML



Course Instructor–
Mrs.Amruta R.Patil
ASSISTANT PROFESSOR

Experiment No.06

Title of Experiment: Program Based on XSLT Stylesheet to Convert XML to HTML

Aim of Experiment: To understand and implement XSLT (Extensible Stylesheet Language Transformations) to transform an XML document into an HTML document.

System Requirements –

Operating System: Windows 10 or above

RAM: Minimum 4 GB

Software/s Needed for Experiment–Web Browser, Text Editor, VSCode

Experiment Objectives:

- To understand the concept of XSLT.
- To learn how XML data is transformed into HTML.
- To write an XSLT stylesheet.
- To link XML with XSLT.
- To display XML data in structured HTML format.

Experiment Outcomes:

- Understand the working of XSLT processor.
- Create an XML document with structured data.
- Write an XSLT stylesheet for transformation.
- Convert XML data into HTML format successfully.

Theory:

XSLT (Extensible Stylesheet Language Transformations) is used to transform XML documents into other formats such as HTML, XML, or plain text.

Program 1: XML File (students.xml)

```
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="students.xsl"?>
<students>
  <student>
    <name>Amruta</name>
    <rollno>101</rollno>
    <branch>Computer Engineering</branch>
  </student>
  <student>
    <name>Rahul</name>
    <rollno>102</rollno>
    <branch>Information Technology</branch>
  </student>
</students>
```

Program 2: XSLT File (students.xsl)

```
<?xml version="1.0"?>
```

```

<xsl:stylesheet version="1.0"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform">
<xsl:template match="/">
<html>
<head>
<title>Student Information</title>
</head>
<body>
<h2>Student Information System</h2>
<table border="1">
<tr>
<th>Name</th>
<th>Roll No</th>
<th>Branch</th>
</tr>
<xsl:for-each select="students/student">
<tr>
<td><xsl:value-of select="name"/></td>
<td><xsl:value-of select="rollno"/></td>
<td><xsl:value-of select="branch"/></td>
</tr>
</xsl:for-each>
</table>
</body>
</html>
</xsl:template>
</xsl:stylesheet>

```

Task for Students: Create an XML document for a **Book Information System** containing the following elements:

<books>, <book>, <title>, <author>, <price>

Task Requirements:

- Create an XML Document
- Create an XSLT file
- Link XML with XSLT
- Use <xsl:for-each>
- Use <xsl:value-of>
- Display data in HTML Table format

Submission Requirement:

- XML File
- XSLT File
- Screenshot of HTML Output
- Screenshot showing transformation working

Observations:

In this experiment, the XML document was successfully created and properly structured with hierarchical elements. The XSLT stylesheet was correctly linked to the XML file using the <?xml-stylesheet?> processing instruction. The XSLT processor applied the template rule and transformed the XML data into a structured HTML format. The <xsl:for-each> element iterated through all the student records, and <xsl:value-of> correctly extracted

and displayed the element values inside an HTML table.

Conclusion:

This experiment demonstrated how XSLT is used to transform XML into HTML format. Students understood how templates and XPath expressions help in extracting and displaying XML data. The transformation was successfully executed using a web browser.

Expected Oral Questions:

1. What is XSLT?
2. What is the purpose of XSLT?
3. What is XPath?
4. What is <xsl:template>?
5. What is <xsl:for-each>?
6. What is <xsl:value-of>?
7. How is XML linked with XSLT?
8. What is an XSLT processor?
9. What is the difference between XML and HTML?
10. Can XSLT transform XML into formats other than HTML?

FAQs in Interview:

1. What is XSLT used for?
2. What is the difference between XSL and XSLT?
3. What is XPath in XSLT?
4. What is template matching in XSLT?
5. What is the use of select attribute?
6. What happens if XPath is incorrect?
7. Can multiple templates be used in XSLT?
8. What is the difference between for-each and apply-templates?
9. Is XSLT case sensitive?
10. What is the namespace used in XSLT?